A B C A L C U L U S

CHAPTER 4

Contextual Applications of Differentiation

AP-Style Practice Questions

EASY MEDIUM HARD

Topics **4.1 – 4.7** AB

Name: _____ Period: _____

PART I

Topics 4.1 – 4.7

Multiple Choice Questions

Show all supporting work on scratch paper. On the AP Exam, Section I is split into a no-calculator and a calculator-allowed part — each question below is labeled accordingly.

Q1 EASY 4.1 Interpreting Derivative No Calculator

$W(t)$ is the weight, in kg, of a calf at age t weeks. The most appropriate units for $W'(t)$ are

(A) kg

(B) weeks

(C) kg/week

(D) weeks/kg

Q2 EASY 4.2 Motion No Calculator

A particle's position is $s(t) = t^2 - 4t$. The particle is at rest when

(A) $t = 0$

(B) $t = 2$

(C) $t = 4$

(D) $t = \pm 2$

Q3 MEDIUM 4.2 Speeding Up No Calculator

A particle moves with velocity $v(t) = t^2 - 4t + 3$. The particle is speeding up on

(A) $(1, 2) \cup (3, \infty)$

(B) $(2, 3)$ only

(C) $(0, 1) \cup (3, \infty)$

(D) $(-\infty, 2)$

Q4 EASY 4.3 Rates (Context) No Calculator

Oil is pumped into a tank so that $V(t) = 10t - t^2/4$ gallons at time t minutes. At $t = 4$, oil enters the tank at

(A) 6 gal/min

(B) 8 gal/min

(C) 10 gal/min

(D) 12 gal/min

Q5 MEDIUM 4.4 Related Rates No Calculator

A spherical balloon is inflated so that its volume increases at 36π cm³/s. At the instant $r = 3$ cm, $\dfrac{dr}{dt} =$

(A) 1 cm/s

(B) 2 cm/s

(C) $\dfrac{1}{3}$ cm/s

(D) 3 cm/s

Q6 MEDIUM 4.4 Related Rates (Shadow) Calculator

A 6-ft-tall person walks away from a 15-ft lamppost at 4 ft/s. The tip of their shadow moves at

(A) $\dfrac{20}{3}$ ft/s

(B) $\dfrac{8}{3}$ ft/s

(C) 6 ft/s

(D) 10 ft/s

Q7 MEDIUM 4.5 Linear Approximation No Calculator

The linear approximation of $f(x) = \sqrt{x}$ at $x = 9$ gives $\sqrt{9.4} \approx$

(A) 3.033

(B) 3.067

(C) 3.100

(D) 3.200

Q8 HARD 4.5 Error Direction No Calculator

Let f be twice-differentiable with $f'' < 0$ on an open interval around $x = a$. The tangent-line approximation to f near $x = a$ is

(A) an overestimate

(B) an underestimate

(C) exact

(D) not determinable from the given information

Q9 EASY 4.7 L'Hôpital No Calculator

$$\lim_{x \to 0} \frac{\sin x}{x} =$$

(A) 0

(B) 1

(C) ∞

(D) does not exist

Q10 MEDIUM 4.7 L'Hôpital No Calculator

$$\lim_{x \to 0} \frac{e^{2x} - 1 - 2x}{x^2} =$$

(A) 0

(B) 1

(C) 2

(D) 4

Q11 MEDIUM 4.7 L'Hôpital (∞/∞) No Calculator

$$\lim_{x \to \infty} \frac{\ln x}{\sqrt{x}} =$$

(A) 0

(B) 1

(C) ∞

(D) does not exist

Q12 HARD 4.4 Related Rates (Angle) Calculator

A spotlight on the ground is 20 m from a wall. A 2-m-tall figure walks from the light toward the wall at 1 m/s. When the figure is 4 m from the wall, the length of the shadow on the wall changes at a rate closest to

(A) -0.31 m/s

(B) -0.63 m/s

(C) $+0.63$ m/s

(D) $+1.25$ m/s

Q13 MEDIUM 4.3 Rates (Economic) No Calculator

A company's cost in dollars for producing x items is $C(x) = 0.01x^2 + 20x + 500$. The marginal cost at $x = 100$ is

(A) 20

(B) 22

(C) 40

(D) 120

Q14 EASY 5.1 MVT Preview of Unit 5 No Calculator

Note: The Mean Value Theorem is formally CED Topic 5.1 (Unit 5). Included here as a preview because tabular-rate FRQs in Unit 4 routinely cite it.

Which hypothesis is required for the Mean Value Theorem on $[a, b]$?

(A) f differentiable on (a, b) and continuous on $[a, b]$.

(B) f continuous on (a, b) .

(C) $f(a) = f(b)$.

(D) f is a polynomial.

Q15 MEDIUM 5.1 MVT Preview of Unit 5 No Calculator

For $f(x) = x^2$ on $[1, 4]$, the value c guaranteed by the MVT is

(A) $\frac{3}{2}$

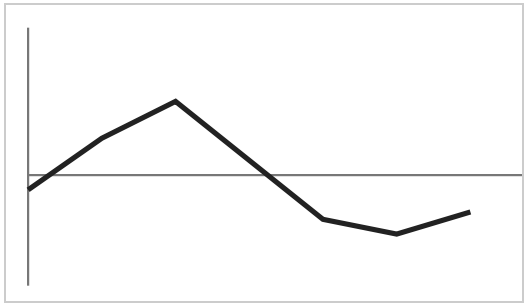
(B) 2

(C) $\frac{5}{2}$

(D) 3

Q16 HARD 4.2 Motion (Graph) No Calculator

A particle's velocity graph is shown on $[0, 6]$. On what interval is the particle speeding up?



(A) $(0, 2)$

(B) $(2, 4)$

(C) $(0, 1) \cup (3, 5)$

(D) $(5, 6)$

Q17 MEDIUM 4.5 Linearization Error No Calculator

$L(x)$ is the tangent-line approximation to $f(x) = \sin x$ at $x = 0$. Then $L(0.1) =$

(A) 0

(B) 0.100

(C) 0.0998

(D) 0.050

Q18 HARD 4.7 L'Hôpital (Twice) No Calculator

$$\lim_{x \to 0} \frac{x - \sin x}{x^3} =$$

(A) 0

(B) $\frac{1}{6}$

(C) $\frac{1}{2}$

(D) 1

PART II *Show All Work*

Free-Response Questions

Contextual FRQs require labeled units, clear statement of what the derivative means in context, and justification using signs of v and a for motion problems. For related-rates, include a diagram, equation relating quantities, and differentiation with respect to time.

FRQ 1 EASY 4.2 Motion No Calculator

A particle moves along the x-axis with velocity $v(t) = 3t^2 - 12t + 9$ for $t \geq 0$ (in m/s).

(a) Find all times $t \geq 0$ when the particle is at rest.

(b) On which interval(s) is the particle moving right?

(c) At $t = 2$, is the particle speeding up or slowing down? Justify using the signs of v and a.

FRQ 2 MEDIUM 4.4 Related Rates Calculator

A water trough is 10 ft long with cross-section an isosceles triangle (2 ft wide at top, 2 ft deep). Water fills the trough at 3 ft³/min.

(a) Express V, the water volume, in terms of the depth h. Justify using similar triangles.

(b) At the instant $h = 1$ ft, find $\frac{dh}{dt}$. Include units.

(c) Is the water surface width increasing faster or slower when $h = 1$ ft vs $h = 1.5$ ft? Justify.

FRQ 3 MEDIUM 4.5 Linearization & Error No Calculator

Let $f(x) = \sqrt[3]{x}$.

(a) Find the linearization $L(x)$ of f at $x = 8$.

(b) Use L to estimate $\sqrt[3]{8.6}$.

(c) Is the estimate an overestimate or underestimate? Justify using f''.

FRQ 4 HARD 4.3 / 4.4 Tabular Rates Calculator

A tank holds $G(t)$ gallons of water at time t minutes. Selected values of G:

t (min)	0	2	4	6	8
$G(t)$ (gal)	120	108	88	60	24

(a) Estimate $G'(4)$ using a symmetric difference quotient. Include units and interpret.

(b) Must there be a time $t \in (0, 8)$ for which $G'(t) = -12$? Justify using the MVT. *(MVT is formally Unit 5, Topic 5.1 — see Q14/Q15 preview above.)*

(c) Use the linearization of G at $t = 4$ to estimate $G(4.5)$. Is the estimate likely an overestimate or underestimate, based on the concavity suggested by the table? Justify.

FRQ 5 HARD 4.7 L'Hôpital & Reasoning No Calculator

Evaluate each limit. For each, state the indeterminate form before applying L'Hôpital's Rule, and justify each step.

(a) $\lim_{x \to 0} \frac{\tan x - x}{x^3}$

(b) $\lim_{x \to \infty} \frac{(\ln x)^2}{x}$

(c) $\lim_{x \to 0^+} x \ln x$

(d) Explain, with a short counterexample, why L'Hôpital's Rule does NOT apply to $\lim_{x \to 0} \frac{x + \sin x}{x}$ as an $\frac{0}{0}$ form — compute the limit directly.